

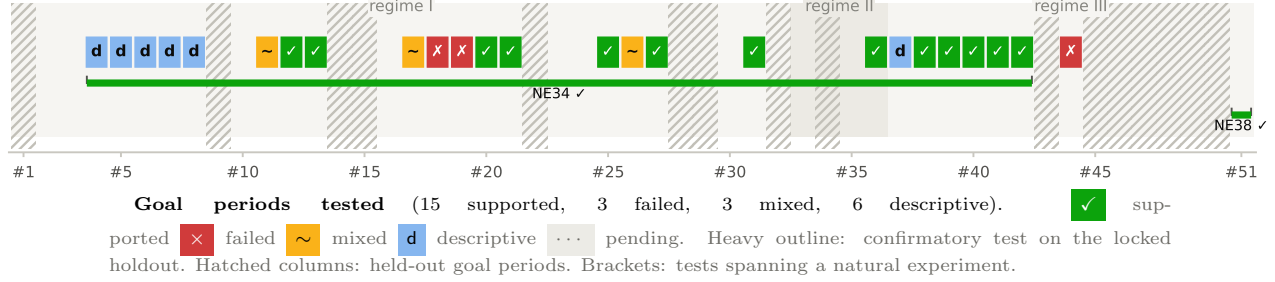
H97: A restoring-force law for the kickoff quench

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 1.5/5

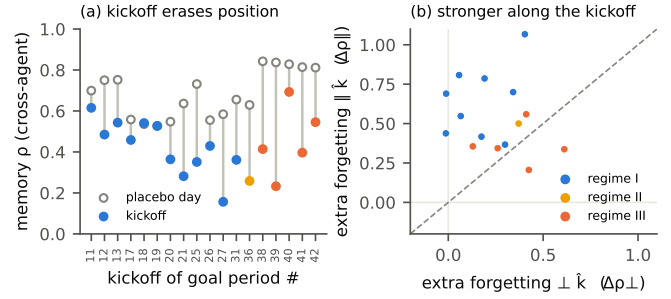
Bottom line. A goal kickoff is a restoring force that erases about a quarter of each agent's relative position beyond an ordinary day, stiffer along the goal and overshooting on day 1; the agent's rate is not shown to be a trait.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH246 says each agent relaxes into the kickoff target at its own rate, $\Delta v_i = \chi_i(\hat{k} - v_i)$. I measured how much of each agent's pre-kickoff position survives day 1, against ordinary days of the same agents, on 18 kickoffs (regimes I–III, 6–15 agents). The synthetic check came first and replaced a biased slope by a scale-free correlation. The kickoff erases position: memory falls from 0.68 to 0.42, $\Delta\rho = +0.26 \pm 0.04$, positive in 16/18 and in all 7 input variants. The law's shape fails. Forgetting is 2–3× larger along $\hat{k}$ than across it, and day 1 overshoots the settled level ($a = +0.12 \pm 0.02$). The per-agent χ_i does not repeat across kickoffs (split-half $r = 0.30$, $p = 0.14$; power 0.30). A one-agent reassignment (NE38) acts as a kickoff on that agent only. Holdout: not run.



Observables. (a) Memory of the pre-kickoff position at each kickoff (filled) and on ordinary days of the same agents (open). The kickoff is below the placebo in 16 of 18. (b) Extra forgetting along the kickoff direction against the transverse one. Points above the diagonal mean a stiffer well along $\hat{k}$; transverse forgetting is still positive (median 0.25), so the kickoff is more than a push along $\hat{k}$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model 11 with overdamped relaxation (model 02). Agent state v_i : mean of unit regime-whitened statement vectors ($d = 32$) on a day. Field: the kickoff, direction $\hat{k}$ from shared `goal_fields`. Memory ρ : cross-agent correlation of positions before and after a boundary, with split-half true variances. ρ_0 : the same on ordinary day boundaries. χ_i : one agent's forgetting of its offset from the others. a : day-1 displacement beyond the law, relative to the days 2–5 level.

$$v_i^{(1)} = t + \beta_i(v_i^{(0)} - t) + \eta_i, \quad \chi_i = 1 - \beta_i$$

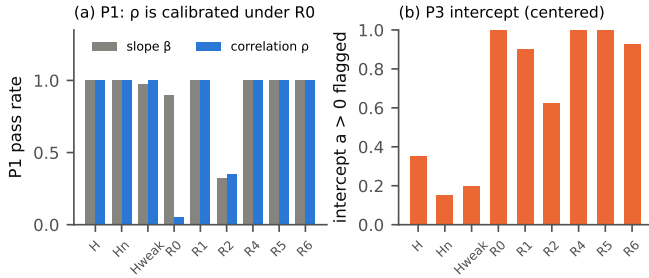
$$\rho = \frac{\text{cov}_i(v^{(1)}, v^{(0)})}{\sqrt{\text{var}_{\text{true}}(v^{(0)}) \text{var}_{\text{true}}(v^{(1)})}}, \quad \Delta\rho = \rho_0 - \rho_{\text{kick}}$$

$$a = \langle \Delta_i \rangle - (1 - \beta_{\parallel}) \langle T - u_i \rangle \quad (= 0 \text{ under the law})$$

Useful as a correction, small as a lever. A kickoff does more than add a common push: it scrambles where each agent was, along the goal and, less, across it. Expect day-1 content to overshoot and then settle back (H54's remanence); do not read the day-1 level as the steady state. The per-agent rate is not a stable trait in these data, so an operator cannot yet rank agents by how fast they adopt a goal. To retarget one agent without disturbing the others, change only its goal text (NE38). For anyone measuring memory in embeddings: a slope between unit-normalized segments fakes forgetting when the common topic grows; use the disattenuated correlation.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\Delta\rho > 0$	$+0.26 \pm 0.04$; 16/18; 7/7 variants	supported
P2 isotropy $\beta_{\parallel} - \beta_{\perp}$	-0.17 ± 0.16 ; style -0.37 ; $\Delta\rho_{\parallel}$ 0.63 vs $\Delta\rho_{\perp}$ 0.25	passes; fragile
P3 intercept $a = 0$	$+0.12 \pm 0.02$ (gte $+0.17$)	failed (R5/R6)
P4 χ_i agent constant	r 0.30 (p 0.14); gte 0.48 (p 0.03); centered -0.06	inconclusive
P6 no lab effect	$p = 0.49$	supported
P8 weekend gap	ρ_0 0.73 vs 0.69	supported
N1 NE38 one agent	χ 0.57/0.70 vs others' 90th pct 0.47/0.40	supported
N2 G44 rooms	$+0.00 / -0.16$	failed
N3 G26 leader's text	below 3/3 placebos; no move toward it	failed (confounded)



Second figure. Synthetic swarms at the real structure of 23 kickoffs, 40 replicates per scenario. (a) A slope-based P1 passes under pure translation (R0) in 90%; the correlation passes in 5%. (b) The centered intercept flags overshoot, common-mode jumps and constant speed (R0, R4–R6) in 90–100%, and the true law in 15–35%.

GOAL PERIODS IN DETAIL

G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G11 $\sim$ mixed
G12 $\checkmark$ supported
G13 $\checkmark$ supported
G17 $\sim$ mixed
G18 $\times$ failed
+17 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The target direction is a text embedding of the kickoff. Day 1 includes quoting the kickoff text. Each kickoff has 6–15 agents, so per-kickoff intervals are wide; the claim rests on the meta-analysis. The isotropy verdict depends on the estimator and the style variant. The previous period's last day can carry wrap-up content that ordinary days lack. Agent constancy has power 0.30, so its miss is not a negative. The G26 test cannot separate the leader's announcement from the day-1 transient.

ROUND 2

Resolve the relaxation in hours after the kickoff, gate it on each agent's read-out call (context ledger), and pool χ_i hierarchically. Run `confirm.py` on the 13 held-out kickoffs.

- **H97-R1** Time-resolved relaxation.
- **H97-R2** Read-out-resolved forgetting.
- **H97-R3** Agent constancy with power.